Priyanshu Tiwari

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Education

Jaypee Institute of Information Technology Sep 2022 – Sep 2026
Bachelor of Technology in Computer Science and Engineering Noida, India

Experience

Software Intern Jun 2026 – Sep 2026
Digiliquid Noida, India (Remote)

- Developed **Python scripts** to clean and format client product data for database migration.
- Tested internal **REST APIs using Postman** and verified JSON payload structures for the development team.
- Resolved frontend UI bugs and updated layout components for internal team dashboards.
- Built a **Python-based NLP pipeline** to analyze customer reviews, categorizing feedback sentiment to help teams identify product issues.

Embedded Systems Intern Jun 2025 – Jul 2025
India Space Lab New Delhi, India (Remote)

- Built a real-time drone tilt detection system on an **ESP32-S3 with an MPU6050**, sampling six axes at **100 Hz** and triggering OLED, LED and buzzer alerts above a **45-degree threshold**.
- Wrote **C++ firmware** using a **complementary filter** (alpha 0.96) and a micros()-measured timestep to combine accelerometer and gyroscope readings.
- Trained a TensorFlow/Keras tilt classifier and compressed it into a **2.5 KB TensorFlow Lite model** for microcontroller flash; simulated the circuit and documented its wiring in Wokwi.

Projects

PolyO ↗

Python, PyTorch, tree-sitter, LightGBM, FastAPI, Docker, GitHub Actions

- Built an ML service that predicts **worst-case time and space complexity** for code in **six languages** without executing it or calling an LLM at inference.
- Normalised code with **tree-sitter** into one shared representation; trained on **194,000 labelled programs** and enforced problem-level splits in CI to prevent leakage.
- Benchmarked four models on a held-out set; the GNN reached **0.377 time and 0.336 space macro-F1**, matching a zero-shot LLM on time and beating it on space at zero API cost. Shipped the model through FastAPI and Docker with a NumPy inference pass verified within 1e-4 of the PyTorch model.

The Aerial Guardian ↗

Python, PyTorch, YOLOv8, OpenCV, ONNX Runtime Web, JavaScript

- Built a real-time drone-video object tracker that runs **entirely in the browser** using **ONNX Runtime Web**, keeping video on the user's device.
- Added a **high-resolution stride-4 detection head** to YOLOv8 for people just **8 to 16 pixels tall** in aerial footage, then retrained it on VisDrone using a 6 GB laptop GPU.
- **Implemented ByteTrack** with Kalman motion prediction and three-stage Hungarian matching to maintain identities through occlusion; used OpenCV to cancel drone-camera motion.

Enagram.io ↗

Next.js, TypeScript, PostgreSQL, pgvector, Gemini, OAuth 2.0

- Built an AI agent that answers questions across a user's **Gmail, Drive and Calendar** in one reply, connecting related people, projects and files.
- Combined **vector embeddings and keyword search** in **PostgreSQL with pgvector** to retrieve both related content and exact strings such as invoice numbers.
- Added **tool-calling** to create and update Calendar events and draft replies in the correct Gmail thread, with per-user access scoped through Google OAuth.

Technical Skills

Languages	Python, C++, JavaScript, SQL, Java
AI and ML	PyTorch, TensorFlow, YOLOv8, OpenCV, scikit-learn, LightGBM, Graph Neural Networks, Computer Vision, Object Detection and Tracking, RAG, Vector Search, LLM APIs (Gemini)
Backend and Data	REST APIs, Node.js, FastAPI, PostgreSQL, pgvector, MongoDB, NumPy, Pandas
Frontend	Next.js, React, Tailwind CSS, HTML5
Developer Tools	Docker, Git, Linux, Postman
CS Core	Data Structures and Algorithms, DBMS, Operating Systems, OOP, Computer Networks

Achievements

- **Competitive Programming: 600+ problems solved** on LeetCode (216 Easy / 317 Medium / **70 Hard**), contest rating **1533**, top **36% globally** across 11 rated contests.

- **Hackathon Finalist:** advanced to the Final Round of **BEcon '24** at **IIT Delhi** among **30+ teams**.